

Scenario - Based Questions - Answers

1. A financial institution plans to deploy a private cloud with strict compliance requirements. Recommend an appropriate cloud datacenter architecture and explain how virtualization and isolation mechanisms can ensure security and regulatory compliance.

Ans: A suitable architecture is a private cloud with multi-tier design. Use separate zones for compute, storage and management. Apply network segmentation using VLANs and firewalls. Virtualization using a type-1 hypervisor enables strong isolation between virtual machines (VMs). Each workload runs in an isolated VM or container, preventing unauthorized access. Use encryption for data at rest and in transit, role-based access control (RBAC), multi-factor authentication (MFA) and audit logging to meet compliance standards like GDPR, PCI-DSS, SOX etc.

2. A SaaS provider experiences inconsistent application performance due to noisy neighbor effects in a virtualized environment. Analyze the root cause and propose resource allocation and hypervisor-level solutions.

Ans: Root cause: Multiple VMs share the same physical resources. If one VM consumes excessive CPU, memory, I/O or network bandwidth, it affects the performance of other VMs.
Solutions: (i) Use resource reservation, limits and quotas.
(ii) Enable CPU and memory scheduling and priorities.
(iii) Use storage and network QoS.
(iv) Implement VM placement policies and live migration.
(v) Monitor resources continuously and auto-scale when needed.

3. An organization wants to migrate legacy monolithic applications into a Service-Oriented Architecture (SOA) in the cloud. Explain the migration strategy and how SOA improves scalability and maintainability.

Ans: Strategy: Analyze the application, identify business functions and break them into independent services. Develop APIs for each service and deploy them in the cloud. Use an API gateway for communication and integrate with other services.
SOA improves scalability because services can be scaled independently. It improves maintainability as updates can be done on individual services without affecting the entire application.

4. A government agency is implementing identity federation across multiple departments using cloud services. Explain how identity management, authentication and privacy policies should be designed to ensure secure access.

Ans: Use federation standards like SAML or OAuth for secure identity sharing. Implement Single Sign-On (SSO) using Identity Provider (IdP). Use Multi-Factor Authentication (MFA) and Role-Based Access Control (RBAC). Define least privilege access and data classification rules. Maintain encryption, audit logs and regular access reviews. Privacy policies should follow data minimization, consent, and compliance with regulations.

5. A cloud provider wants to implement real-time user presence detection for collaborative applications. Discuss how presence services can be integrated into cloud architecture while maintaining user privacy and system scalability.

Ans: Use a presence service built on publish/subscribe model using message brokers (e.g., MQTT, XMPP). Store presence data in distributed in-memory databases for low latency. Integrate with application servers via APIs or SDKs. Ensure privacy by sharing only necessary status, using user consent and encryption. For scalability, use auto-scaling instances, load balancing and geo-distributed architecture.



SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 25

Question Count: 5

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Criteria	Understanding of Scenario (1)	Identification of Risks / Issues (1)	Application of Concepts (1.25)	Decision Justification (1)	Clarity (075)	TOTAL (5)
Weightage	20%	20%	25%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded: